Umang Singh

Software engineer with **2+ years at Oracle** building cloud systems, **multiple hackathon wins**, and 3 peer-reviewed papers.
Portfolio | LinkedIn | GitHub | umang2singh@gmail.com | +91 9561537119 | India

Technical Skills

Languages & Backend: Java, Python, SQL, Kotlin, Dart, Spring Boot, FastAPI, REST APIs, asyncio, Postgres, Flutter
Cloud Systems: GCP (Vertex AI, GKE, Cloud Run), AWS (Bedrock, Lambda, S3), Oracle Cloud Infrastructure
DevOps & Observability: Docker, Kubernetes, Helm, Terraform, Jenkins, GitHub Actions, CI/CD, distributed tracing
AI/ML: Agentic AI (LangGraph), MCP, RAG (pgvector), fine-tuning, PyTorch, Hugging Face, guardrails, evaluation harnesses

Work Experience

Oracle — Mumbai, India
Software Engineer 2 — Jul 2024 – Present

- Built **provisioning and FastAPI microservices** for an anti-money-laundering SaaS and migrated metering from Spring Boot to Helidon, cutting startup **~40%** and memory **~25%**. Collaborated across **6 teams** to launch it to customers with **zero outages**.
- Improved the reliability of an ML batch job so it recovers on its own from **database outages of up to 15 minutes** and **Kubernetes node replacement**, using database-request replay and idempotent restarts.
- Achieved easier maintenance and testing by splitting a **30,000+ line Python monolith (200+ APIs)** into separate analytics and data-access layers in about **3 weeks** instead of an estimated **2 months**, using AI-assisted development.

Project Intern — Jun 2023 – Jul 2023

- Built an automated data-integration pipeline with **150+ field mappings**, bringing customer, account and transaction data together into a unified case-management system for financial-crime investigations.

Futurepreneur Tech — Mumbai, India
Technical Lead — Feb 2023 – May 2023

- Shipped a hostel-discovery app on **Flutter** and **Firebase** with optimistic UI and onboarded **20+ hostels**.
- Secured **INR 100,000** in funding from the college principal and led a **15+ person team** across UI/UX, frontend and backend.

Achievements

- **1st Runner-Up** at the Google Agentic AI Hackathon 2025 (**57,000+** registrants) and the Japan-India Hackathon 2024.
- **Winner** of Smart India Hackathon (SIH) 2023, run by the Government of India with **250,000+** participants.
- **Finalist at national hackathons**: BIAL Genie by **Microsoft** (top 8 of **11,000+**) and ONDC-Google Build for Bharat.
- Ranked **1,081 of 24,578** (**top 4.4%**) in LeetCode Weekly Contest 352, with **700+** problems solved across 500+ active days.

Projects

***ChronoTrace*: AI Agent for Flaky-Test Repair** | Python, asyncio, Amazon Bedrock, Strands Agents — Aug 2026 – Sep 2026

- Built a **flaky-test AI agent** optimized for race conditions in concurrent code that finds the steps running out of order and fixes their ordering, while the same LLM alone masked the race with a sleep or retry in **86%** of cases

***KrishiBodha*: Agentic AI Assistant for Farmers** | Gemini, Gemma, Vertex AI, Google ADK, Flutter — Jun 2025 – Aug 2025

- **Led the team** behind a multilingual voice assistant for farmers covering crop disease, market alerts, subsidies, and crop and soil health from **Sentinel-2** satellite imagery, running **11 AI agents** on Gemini with an **on-device Gemma** model for offline use

***VidyuRakshak*: Power-Line Safety from Satellite Imagery** | Google Earth Engine, Random Forest, AWS — Dec 2023 – Aug 2024

- Trained a **Random Forest** model that estimates tree height from 7 vegetation indices in satellite imagery so power utilities can spot trees growing too close to their lines, validated against elevation data on **107,000** records
- After it won SIH 2023, Power Grid Corporation of India signed an agreement to develop it further, and the team kept the IP

Publications

- "*Advanced Deep Learning Techniques for Early Detection of Oral Cancer: A Multimodal Approach*"; *ISAI, Springer.* — *Mar 2025*
- "*Exploring Challenges and Results in Crop Type Mapping Using Satellite Imagery*"; *IEEE InGARSS. First author.* — *Dec 2024*
- "*Detecting Tree Height and Addressing Challenges Along Power Line Corridors...*"; *ICICBDA, Springer.* — *Jun 2024*

Leadership and Extracurriculars

- **Mentored 15 AI/ML teams** at HackSync 2.0, TSEC's hackathon, on problem framing and model choice.
- **Conducted a database workshop** for **100+ students** at a TSEC alumni session on connecting Java, Python and MERN apps.

Education

University of Mumbai | College Hall of Fame Award (**30 of 600+** students) — Mumbai, India
Bachelor of Engineering, Computer Engineering | **GPA: 8.76/10** — 2020 – 2024